

AHTARI MYLLYMAKI, Finland

WMO: 029240

Lat: 62.535N

Lon: 24.217E

Elev: 524

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.8	-11.7	-23.0	1.6	-17.7	-16.3	2.3	-11.5	9.7	23.7	8.5	22.9	0.4	40	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	78.8	63.5	75.3	61.9	72.1	60.4	66.5	74.3	64.3	71.4	62.3	69.4	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	90.2	69.4	61.5	83.3	67.4	59.5	77.4	65.1	31.4	74.3	29.7	71.4	28.3	69.3	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.7	7.1	-22.6	82.9	7.5	3.7	-28.0	85.6	-32.4	87.8	-36.5	89.9	-42.0	92.6		
(5)			-23.0	69.4	7.3	3.1	-28.2	71.6	-32.5	73.5	-36.5	75.2	-41.8	77.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	17.7	17.5	24.4	36.0	46.3	54.9	60.0	57.1	48.1	38.1	30.1	23.6		
	DBStd	17.08	12.67	13.08	9.59	6.47	7.48	6.27	5.41	5.41	6.36	7.31	8.40	11.70		
	HDD50	5214	1000	909	793	421	163	22	1	7	110	372	598	818		
	HDD65	9909	1465	1329	1258	870	581	310	174	249	507	835	1048	1283		
	CDD50	811	0	0	0	1	47	168	311	229	53	2	0	0		
	CDD65	30	0	0	0	0	0	5	19	6	0	0	0	0		
	CDH74	419	0	0	0	0	29	91	235	64	0	0	0	0		
	CDH80	52	0	0	0	0	1	9	38	4	0	0	0	0		
	WSAvg	2.9	2.7	2.7	3.1	3.0	3.2	3.0	2.6	2.5	2.7	3.1	3.2	3.0		
	PrecAvg	24.1	1.7	1.3	1.2	1.3	1.7	2.6	3.3	2.9	2.2	2.3	1.9	1.7		
(15) Precipitation	PrecMax	32.0	3.5	2.6	1.9	2.6	3.8	4.6	5.8	6.4	5.4	4.8	3.7	3.2		
	PrecMin	18.5	0.4	0.0	0.1	0.2	0.3	0.8	0.8	0.9	0.6	0.5	0.1	0.6		
	PrecStd	3.3	0.8	0.7	0.5	0.7	0.9	1.0	1.3	1.1	1.3	1.2	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	40.4	48.9	64.0	77.8	81.6	84.2	79.2	68.5	55.9	48.2	43.0		
		MCWB	36.5	36.5	40.4	49.8	59.0	63.6	67.6	64.8	59.6	51.2	47.6	42.1		
	2%	DB	36.0	36.9	43.1	57.1	72.2	76.7	80.0	75.5	64.8	52.7	45.2	40.3		
		MCWB	34.9	35.1	36.8	45.8	56.2	61.1	64.9	63.0	57.6	50.6	44.0	39.5		
	5%	DB	34.3	34.6	39.5	52.6	67.1	72.5	76.4	72.2	61.5	50.8	42.9	38.0		
		MCWB	33.5	33.6	34.9	43.2	53.6	58.7	63.2	61.7	55.8	48.9	42.0	37.0		
	10%	DB	32.7	32.8	36.6	48.6	62.1	68.9	72.8	69.0	58.8	48.8	40.5	35.8		
		MCWB	32.2	32.0	33.3	41.2	50.8	57.2	62.0	60.3	54.6	47.1	39.7	35.1		
	0.4%	WB	37.1	37.2	41.2	51.6	61.2	66.4	70.2	68.2	61.2	52.9	47.3	42.2		
		MCDB	37.7	39.1	47.5	58.8	71.8	76.9	80.4	75.4	66.2	54.2	48.0	42.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.1	35.5	37.9	47.6	58.2	63.2	67.5	65.2	58.9	51.1	44.1	39.7		
		MCDB	35.8	36.8	41.6	55.5	68.0	72.2	75.5	71.8	62.7	52.3	44.8	40.1		
	5%	WB	33.7	33.8	35.8	44.4	55.1	60.8	65.5	63.2	57.1	49.2	42.1	37.1		
		MCDB	34.2	34.6	38.6	51.5	65.1	68.9	72.9	69.7	60.5	50.6	42.8	37.7		
	10%	WB	32.3	32.3	33.8	41.7	51.9	58.7	63.4	61.4	55.3	47.3	39.8	35.1		
		MCDB	32.8	32.9	35.9	47.7	59.8	66.7	70.4	67.2	58.1	48.6	40.5	35.7		
	5% DB	MDBR	11.3	13.5	17.5	18.4	21.2	19.3	19.0	18.9	15.9	9.9	7.1	9.1		
		MCDBR	8.1	8.8	19.2	26.9	31.3	27.1	25.8	25.1	19.9	11.6	7.3	7.0		
		MCWBR	7.9	7.8	13.8	16.4	16.6	13.5	12.3	13.5	13.0	9.2	7.1	6.8		
		MCWBR	8.1	8.5	15.8	24.6	27.7	21.9	21.6	20.8	16.9	10.3	7.3	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.286	0.312	0.348	0.344	0.344	0.367	0.352	0.328	0.313	0.254	0.355		
		taud	2.042	2.148	2.215	2.360	2.459	2.482	2.435	2.467	2.527	2.404	2.141	2.624		
	Ebn at Noon		133	207	249	266	278	279	268	263	249	202	125	81		
		Edh at Noon	11	21	29	31	30	30	31	28	22	17	10	6		
(44) All-Sky Solar Radiation	RadAvg		60	236	707	1156	1544	1663	1577	1214	705	283	73	30		
	RadStd		9	56	78	140	170	138	161	134	85	56	14	3		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	-243	N/A	N/A	N/A

Nomenclature: See separate page